SUBASH K

MECHANICAL ENGINEER

9150507057 | subashkasi36@gmail.com | linkedin.com/in/subashk280999 | Salem, TamilNadu, India

PROFESSIONAL SUMMARY

Mechanical Engineering graduate with a strong core-engineering foundation, including a hands-on internship at BMW, who transitioned into software development. Full-Stack Developer with 4 years of experience building scalable web and mobile applications using React, Flutter, Firebase, and Google Cloud Platform. Proven ability to design, develop, and deploy end-to-end SaaS products independently. Experienced in developing RESTful APIs, real-time systems, and serverless architectures, including LLM-powered features built with RAG pipelines and vector databases. Strong in backend system design, database optimization, and AI-augmented development to accelerate delivery while maintaining high-quality production standards.

TECHNICAL SKILLS

CAD & Design: AutoCAD, CATIA (working knowledge / academic exposure)

Manufacturing: CNC Machining, Lathe & Milling Operations, Sheet Metal, Welding

Core Engineering: Thermodynamics, Fluid Mechanics, Strength of Materials, Machine Design

Frontend: React.js, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS, Material UI

Mobile: Flutter, Dart, Cross-Platform Development (Android & iOS)

Backend: Node.js, REST APIs, Firebase Cloud Functions, Serverless Architecture

Cloud & Database: Firebase (Firestore, Realtime DB, Authentication), Google Cloud Platform, NoSQL, SQL (Basic)

AI Tools: Cursor IDE, Claude AI, Prompt Engineering, AI-Assisted Development

PROFESSIONAL EXPERIENCE

Gen AI Engineer — DXC Technology, Bengaluru *June 2022 – Sep 2024*

- Analyzed enterprise application behavior for Allianz's Command Workstation platform, applying AI-assisted analysis to identify functional and performance issues
- Identified, classified, and resolved P1/P2/P3 production issues in consultation with 10+ developers and Tech Support team, qualifying 50+ patches across field defects, product updates and feature requests
- Used LLM-based AI tools to accelerate script generation, log analysis, and defect root-cause investigation, reducing turnaround time on issue resolution
- Built automation and validation scripts for Client-Server applications integrated with Citrix receiver server, applying prompt-driven AI workflows to generate and refine scripts faster
- Collaborated with a 10-member Agile team and mentored new joiners in Domain Knowledge and AI-assisted workflows, developing structured prompt libraries and AI-assisted documentation for requirement traceability

Key Project: Allianz Project (Client – Allianz) | Role – Gen AI Engineer (2.3 years) | Technologies – Command Workstation, LLM-Assisted Automation, Prompt Engineering

AI-Augmented Full-Stack Developer — Ajantha Tech Controls, Salem, TamilNadu, India *Oct 2024 – Present*

Project: School Management System (SaaS – Web & Mobile)

- Designed and developed an AI-augmented SaaS platform handling academic, administrative, and financial workflows
- Built 5 role-based dashboards (Admin, Teacher, Student, Finance, Super Admin) with secure authentication and RBAC using AI-assisted code generation
- Developed cross-platform mobile apps (Android & iOS) in Flutter from a single codebase, accelerating UI development through AI pair-programming
- Implemented real-time features including GPS bus tracking, attendance, exam results, and fee management
- Created 8+ core modules including student management, reports, transport, and hostel systems using prompt-driven development workflows
- Built scalable serverless backend with Firebase Cloud Functions, applying AI-assisted API design and code review
- Optimized database queries with AI-driven analysis, reducing cloud costs by ~25% while maintaining high performance
- Managed complete product lifecycle from requirements to deployment and production support

- Accelerated delivery using Claude AI, Cursor IDE, and multi-agent AI workflows, reducing feature development time by 30–40%

PROJECTS

Automated Guided Vehicle (AGV) with Lifting Trolley — Final-Year Mechanical Engineering Project

- Designed and fabricated an Automated Guided Vehicle capable of autonomously transporting material loads within a shop-floor environment
- Integrated a mechanical lifting trolley mechanism to raise, carry, and lower loads, reducing manual material-handling effort
- Developed the chassis, drive system, and lifting assembly, applying core concepts of machine design, mechanisms, and structural strength
- Implemented line-following/path-guided navigation with sensor-based control for automated movement along a predefined route

KEY ACHIEVEMENTS

- Delivered a production-grade SaaS application as a solo developer
- Reduced cloud infrastructure costs through efficient database and query optimization
- Built and deployed cross-platform mobile apps from a single codebase
- Achieved team-level productivity using AI-assisted development workflows
- Successfully transitioned from Mechanical Engineering to Software Development

CERTIFICATION

ISTQB Certified Tester – Foundation Level (CTFL)

Indian Testing Board (ISTQB) | March 2023 | Credential ID: CTFL-121844

ACADEMIC BACKGROUND

B.E. Mechanical Engineering — Knowledge Institute of Technology (Anna University)

2018 – 2022 | CGPA: 8.0

CONTINUOUS LEARNING

- Active practitioner of AI-augmented development workflows using Cursor IDE, Claude Code CLI, and multi-agent orchestration for production software delivery
- Experienced in prompt engineering and MCP server configuration for connecting AI agents to external tools, databases, APIs, and testing workflows
- Self-taught full-stack developer demonstrating rapid learning, adaptability, and strong problem-solving across emerging technologies

ADDITIONAL SKILLS

Languages: English (Fluent), Tamil (Native)

Communication: Public Speaking, Report Creation, Presentation Skills, Cross-functional Collaboration

Interpersonal: Teamwork, Leadership, Conflict Resolution, Empathy, Active Listening

Soft Skills: Adaptability, Problem Solving, Creativity, Critical Thinking, Attention to Detail

Availability: Open for onsite assignments (short or long-term); willing to relocate

---

Date: 08-Jul-2026 Place: Salem **Subash K**